Curriculum Vitae

IVAN B. PENKOV

Personal — US citizen, born May 19, 1957, in Sofia, Bulgaria, wife Irina Sousokolova, son Boyan and daughter Anna

University Education

June 1982 — Master Degree in Mathematics
Moscow State University
Thesis: The inverse Penrose transform
Adviser: Professor Yu. I. Manin

Oct. 1987 — Candidate of Physical and Mathematical Sciences (Ph.D.),
Steklov Mathematical Institute
Thesis: Typical modules over classical Lie superalgebras and their geometric realizations
Adviser: Professor Yu. I. Manin
Official referees: Professors D. Zhelobenko and A. Rudakov

Fellowships and Awards

1991–1995 — NSF research grant (jointly with R. E. Block and V. Chari).

1995 — NSF grant for organizing the conference in honor of R. E. Block (jointly with V. Chari).

1995 — Grant from the Erwin Schrödinger Institute for the program "Representation Theory with Applications to Mathematical Physics" (jointly with Joseph A. Wolf).

1996 — NSF grant for support of US graduate students at the Erwin Schrödinger Institute.

1995–1997 — NSF research grant.

1997–2002 — NSF Group Infrastructure Grant "Lie Groups, Lie Algebras and Their Representations" (jointly with Joseph A. Wolf and Geoffrey Mason).

Jan–May, 2002 — Coorganizer of the one-semester MSRI Program "Representations of Infinite Dimensional Algebras and Mathematical Physics" (jointly with Edward Frenkel, Victor Kac, Vera Serganova and Gregg Zuckerman).

April 2002 — Coorganizer of the MSRI Workshop "Quantum field theory and supersymmetry".

Sept. 2003	Coorganizer of the workshop "Locally finite Lie algebras at the Banff Research Institute, Canada.
2003–2006	NSF Grant "Lie Groups, Lie Algebras and Their Representations" (jointly with Joseph A. Wolf and Geoffrey Mason).
2009–2012	DFG grant (one postdoc position).
2009	Volkswagen Foundation grant for organizing the summer school "Structures in Lie Representation Theory" (jointly with A. Joseph and A. Melnikov).
2009-2012	DFG grant within the Schwerpunktprogramm 1388 "Darstellungstheorie".
2012-2013	DFG grant for German-Israeli cooperation.
2012-2015	DFG grant within the Schwerpunktprogramm 1388 "Darstellungstheorie".
2014	Coorganizer of the annual meeting in Soltau of the priority program "SPP 1388 Darstellungstheorie".
2014	Coorganizer of the meeting "Hodge theory and unitary representations of reductive Lie groups", Jacobs University Bremen.
2015	Coorganizer of the meeting "Groups and Rings Theory and Applications", July 15-22, Sofia, Bulgaria.
2015-2016	AIM SQuaRE grant
2015–2017	DFG grant (3/4 postdoc position)
2017	Vera Serganova and Joseph A. Wolf organized a two-day workshop on the occasion of my 60th birthday at UC Berkeley on May 13-14.
2018-2020	DFG grant (one postdoc position)

Professional Experience

1982–83	Research Assistant, Institute of Mathematics Bulgarian Academy of Sciences, Sofia
May–June 1986	Visiting Professor at Institute Fourier, Grenoble, France
1987–89	Research Fellow, Section of Algebra, Institute of Mathematics, Bulgarian Academy of Sciences, Sofia, Bulgaria

Oct. 1989	Visiting Professor at the University of Paris VII, France
Jan.–Dec. 1990	Visiting Assistant Professor, Department of Mathematics, University of California at Berkeley
June–July 1990	Visiting Scholar at the University of Göttingen, Germany
Jan.–June 1991	Visiting Assistant Professor, Department of Mathematics, University of California at Davis
July 1991–June 1993	Assistant Professor, Department of Mathematics, University of California at Riverside
Sept.–Oct. 1993	Visiting Scholar at the Erwin Schrödinger Institute, Vienna, Austria
July 1993–June 2000	Associate Professor (tenured appointment), Department of Mathematics, University of California at Riverside
Apr.–May 1996	Visiting Scholar and Program Organizer at The Erwin Schrödinger Institute, Vienna, Austria
June 1996	Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany
June 1997	Visiting Professor at the University of Burgundy Dijon, France
Sept. 1997	Visiting Professor at the University Louis Pasteur, Strasbourg, France
Aug.–Sept. 1998	Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany
Nov. 1999	Visiting Professor at the University of Hamburg, Germany
April-June 2000	Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany
July 2000 – Oct 2004	Professor (tenured appointment), Department of Mathematics, University of California at Riverside
Aug.–Sept. 2001	Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany
Jan.–March 2002	Program organizer and MSRI scholar, MSRI, Berkeley
Aug.–Sept. 2002	Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany

Fall 2003	Visiting Professor at Yale University
Spring 2004	Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany
Fall 2007	Visiting Professor at University of California at Berkeley
Summer 2008	Visiting Professor at the University of Sao Paulo, Brazil
Spring 2012	Visiting Professor at Yale University
Fall 2012	Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany
Fall 2004–present	Professor, Jacobs University Bremen, Germany (named International University Bremen before 2007)

Main Research Interests

Representations of finite and infinite dimensional Lie algebras, generalized Harish-Chandra modules, representations of Lie superalgebras and Lie supergroups, geometry of supermanifolds, homogeneous superspaces, super D-modules, homogeneous ind-spaces and geometry of ind-varieties.

Other Professional Activities

1991–2006	Founding coorganizer of the California Lie Theory Program, a cooperation program in Lie Theory between the campuses of the University of California.
2004	Founding coorganizer of the Program "Darstellungstheorie, Transformationsgruppen und Mathematische Physik" of the consortium of five universities: IU-Bremen and the Universities of Bochum, Darmstadt, Hamburg, and Köln.
2005	Coorganizer of the conference in honor of Professor Yu. I. Manin at the Max Planck Institute for Mathematics, February 2005. Coorganizer of the International Conference in Honor of R.O. Wells, Jr., IUB, December 1-2, 2005.
2010	Coorganizer of the International Conference in "Infinite-dimensional Lie Algebras" in Newfoundland, Canada.
2011	Coorganizer of the conference "Lie Groups" in honor of J.A. Wolf, Bochum, Germany.

2012	Coorganizer of the German-Israeli workshop "Symplectic Geometry with applications to Representation Theory" at Jacobs University, Bremen, Germany. Organizer of the meeting "7 1/2" in honor of Yu. I. Manin at IHES, Bures sur Yvette, France.
2013	Organzier of the meeting "Geometry" in honor of Vasil Tsanov, Sofia, Bulgaria.
2014	Coorganizer of the annual meeting in Soltau of the priority program "SPP 1388 Darstellungstheorie". Coorganizer of the meeting "Hodge theory and unitary representations of reductive Lie groups", Jacobs University Bremen.
2019	Coorganizer of the meeting "Perspectives on Representations of Semi-Simple Lie algebras" in honor of Gregg Zuckerman on the occasion of his 70-th birthday.

List of Publications
IVAN B. PENKOV

Research papers

1. The Penrose transform on general grassmannians, C.R. Acad. Sci. Bulg., 33 (1980), 1439-1442.

2. Null-geodesics of complex Einstein spaces, Funk. Anal. i Priloz., 16, No.1 (1982), 78-79,(Russian), (with Yu.I.Manin).

3. Linear differential operators and cohomology of analytic spaces, Uspekhi Mat. Nauk, 37, No 4 (1982), 171-172, (Russian).

4. D-modules on supermanifolds, Invent.Math., 71 (1983), 501-512.

5. The formalism of left and right connections on supermanifolds, (1982), Lect. on supermanifolds, geom. methods and conf. groups, editor: Döbner et al, World Scientific, Singapore, 1989, 3-13 (with Yu. I. Manin).

6. An introduction to geometric representation theory for complex simple Lie superalgebras, Proc. Conf. Diff. Geom. Math. in Physics, Shumen 1984, World Scientific 1986, 89-106.

7. Serre duality for projective supermanifolds, Funk. Anal. i Priloz., 18,No.1 (1984), 78-79, (Russian) (with O. Ogievetsky).

8. Projectivity and D-affinity of flag supermanifolds, Uspekhi Math. Nauk, 40 No.1 (1985), 211-212, (Russian) (with I. Skornyakov).

9. A geometric approach to the linear Penrose transform, Transactions AMS, 290 (1985), 555-575.

10. Cohomologie des D-modules typiques sur les supervariétés de drapeaux, C. R.-Acad. Sci., Paris, 199 (1984), 1005-1008 (with I. Skornyakov).

11. Characters of typical irreducible finite dimensional $q(m)$-modules, Funk. Anal. i Priloz., 20 (1986), No. 1, 37-45 (Russian).

12. Localisation des representations typiques d'une superalgèbre de Lie complexe classique, C. R. Acad. Sci. Paris, 304 (1987), 163-166.

13. Borel-Weil-Bott theory for classical Lie supergroups, Curr. Probl. Math., Moscow, 32 (1988), 71-124 (Russian).

14. Elements of supergeometry, Curr. Probl. Math., Moscow, 32 (1988), 3-25, (Russian), (with Yu. I. Manin and A. Voronov).

15. Character formulas for some classes of atypical $gl(m + n\varepsilon)$-and $p(m)$-modules, Lett. Math. Phys., 16 (1988), 251-261 (with V. Serganova).

16. Cohomology of G/P for classical complex Lie supergroups G and characters of some atypical G-modules, Annales Fourier, 39 (1989), 845-873, (with V. Serganova).

17. Generic irreducible representations of classical Lie superalgebras, Lecture Notes in Physics, 375 (1991), Springer Verlag, Berlin, 311-319 (with V. Serganova).

18. Representations of classical Lie superalgebras of type 1, Indag. Math., N.S. 3(2), (1992), 419-466 (with V. Serganova).

19. Generic representations of classical Lie superalgebras and their localisation, Monatsh. Math., 118 (1994), 267-313.

20. Generic irreducible representations of finite-dimensional Lie superalgebras, Intern. Journ. Math., 5 (1994), 389-419 (with V. Serganova).

21. Partially and fully integrable modules over Lie superalgebras, Studies in Advanced Mathematics (series editor S.-T. Yau), 4, AMS and Internatl. Press 1997, 49-67 (with I. Dimitrov).

22. Characters of irreducible G-modules and cohomology of G/P for the Lie supergroup $G = Q(n)$, Journ. of Math. Sci. 84 (1997), 1382-1412 (with V. Serganova).

23. Characters of finite-dimensional irreducible $q(n)$-modules, Lett. Math. Phys. 40 (1997), 147-158 (with V. Serganova).

24. Characters of strongly generic irreducible Lie superalgebra representations, Intern. Journ. Math. 9 (1998), 331-366.

25. Partially integrable highest weight modules, Transformation Groups 3 (1998), 241-253 (with I. Dimitrov).

26. The support of an irreducible Lie algebra representation, Journ. of Algebra 209 (1998), 129-142 (with V. Serganova).

27. Weight modules of direct limit Lie algebras, IMRN 1999, no. 5, 223-249 (with I. Dimitrov).

28. Weight representations of the Cartan type Lie algebras W_n and $\bar{S}_n$, Math. Research Lett. 6 (1999), 397-416. (with V. Serganova).

29. On the structure of weight modules, Transactions AMS 352 (2000), pp. 2857-2869 (with I. Dimitrov and O. Mathieu).

30. Generalized Harish-Chandra modules, Moscow Math. Journal 2 (2002), 753-767 (with V. Serganova).

31. A Bott-Borel-Weil theory for direct limits of algebraic groups, American Journ. of Math. 124 (2002), 955-998 (with I. Dimitrov and J. A. Wolf).

32. Locally finite Lie algebras with root decomposition, Archiv der Math. 80 (2003) 478-485 (with H. Strade).

33. Cartan subalgebras of gl_∞, Canadian Math. Bull., 46 (4), 2003, 597-616 (with K.-H. Neeb).

34. Finite rank vector bundles on inductive limits of grassmannians, IMRN 2003, no. 34, 1871-1887 (with J. Donin).

35. Classically semisimple locally finite Lie superalgebras, Forum Math. 16 (2004), 431-446.

36. On the existence of (g,k)-modules of finite type, Duke Math. Journal 125 (2004), 329-349 (with V. Serganova and G. Zuckerman).

37. A Reduction Theorem for Highest Weight Modules over Toroidal Lie Algebras, Comm. Math. Physics 250(2004), no. 1, 47-53 (with I. Dimitrov and V. Futorny).

38. Generalized Harish-Chandra modules: a new direction of the structure theory of representations, Acta Applicandae Mathematicae, 81(2004), 311-326(with G. Zuckerman).

39. Ind-varieties of generalized flags as homogeneous spaces for classical ind-groups, IMRN 2004, no. 55, 2935-2953(with I. Dimitrov).

40. Borel subalgebras of $gl(\infty)$, Resenhas IME-USP 6 (2004), 111-119(with I. Dimitrov).

41. Generalized Harish-Chandra modules with generic minimal k-type, Asian Journal of Mathematics 8(2004), 795-812(with G. Zuckerman).

42. A construction of generalized Harish-Chandra modules with arbitrary minimal k-type, Canad. Math. Bull. 50 (2007), 603-609 (with G. Zuckerman).

43. Cartan subalgebras of root-reductive Lie algebras, Journal of Algebra, 308 (2007), 583-611 (with E. Dan-Cohen and N. Snyder).

44. A construction of generalized Harish-Chandra modules for locally reductive Lie Algebras, Transformation Groups 13 (2008), 799-817 (with G. Zuckerman).

45. Locally semisimple and maximal subalgebras of the finitary Lie algebras $gl(\infty)$, $sl(\infty)$, $so(\infty)$, and $sp(\infty)$, Journal of Algebra 322 (2009), 2069-2081 (with I. Dimitrov).

46. Parabolic and Levi subalgebras of finitary Lie algebras, IMRN, International Mathematics Research Notices, Article ID rnp169, doi:10.1093/imrn/rnp169 (with E. Dan-Cohen).

47. Parabolic Subgroups of Real Direct Limit Lie Groups, Contemporary Mathematics 499, 2009, pp. 47-59 (with E. Dan-Cohen and J. A. Wolf).

48. Rank 2 vector bundles on ind-grassmannians, Manin Festschrift,in Progress in Mathematics 270, Birkhaeuser Boston, 2009, pp. 555-572 (with A. S. Tikhomirov).

49. Bounded simple (g; sl(2))-modules for rkg = 2, Journal of Lie Theory 20 (2010), 581-615 (with V. Serganova).

50. Tensor representations of infinite-dimensional root-reductive Lie algebras, in Developments and Trends in Infinite-Dimensional Lie Theory, Progress in Mathematics 288, Birkhäuser, 2011, pp. 127-150 (with K. Styrkas).

51. On bounded generalized Harish-Chandra modules, Annales de l'Institut Fourier, 62 (2012), 477-496 (with V. Serganova).

52. Triviality of vector bundles on twisted ind-Grassmannians, Mat. Sbornik 202:1 (2011), 1-39 (with A. S. Tikhomirov).

53. A Bott-Borel-Weil theorem for diagonal ind-groups, Canadian Journal of Mathematics, doi:10.4153/CJM-2011-032-6 (with I. Dimitrov).

54. Categories of integrable $sl(\infty)$-, $o(\infty)$-, $sp(\infty)$-modules, in "Representation Theory and Mathematical Physics", Contemporary Mathematics 557, AMS 2011, pp. 335-357 (with V. Serganova).

55. Levi components of parabolic subalgebras of finitary Lie algebras, in "Representation Theory and Mathematical Physics", Contemporary Mathematics 557, AMS 2011, pp. 129-149 (with E. Dan-Cohen).

56. A Koszul category of representations of finitary Lie algebras, Advances of Mathematics 289 (2016), 250-278 (with Elizabeth Dan-Cohen and Vera Serganova).

57. On the structure of the fundamental series of generalized Harish-Chandra modules, Asian Journal of Mathematics , 16(2012), 489-514 (with G. Zuckerman).

58. On Ideals in the enveloping algebra of a locally simple Lie algebra, Int Math Res Notices (2015) Vol. 2015 5196-5228 first published online June 10, 2014 doi:10.1093/imrn/rnu085 (with Alexey Petukhov).

59. Linear Ind-Grassmannians, Pure and Applied Math. Quarterly 10 (2014), 289-323 (with Alexander Tikhomirov).

60. Algebraic methods in the theory of generalized Harish-Chandra modules, Developments and Retrospectives in Lie Theory: Algebraic Methods, Developments in Mathematics, vol. 38, Springer Verlag, pp. 331-350 (with Gregg Zuckerman).

61. Tensor representations of Mackey Lie algebras and their dense subalgebras, Developments and Retrospectives in Lie Theory: Algebraic Methods, Developments in Mathematics, vol. 38, Springer Verlag, pp. 291-330 (with Vera Serganova).

62. On the Barth-Van de Ven-Tyurin-Sato Theorem, Sbornik: Mathematics, **206**:6 (2015), 814-848 (with Alexander Tikhomirov).

63. A categorification of the Boson-Fermion correspondence via Representation Theory of $sl(\infty)$, Comm. Math. Phys. **341**:3 (2016), pp. 911-931 (with Igor Frenkel and Vera Serganova).

64. Annihilators of highest weight $sl(\infty)$-modules, Transformation Groups 21 (2016), 821-849 (with A. Petukhov)

65. Infinite Konstant cascades and centrally generated primitive ideals of $U(n)$ in types A_∞, C_∞, Journal of Algebra **447**(2016), 109-134 (with Mikhail Ignatyev).

66. Schubert decompositions for ind-varieties of generalized flags, Asian Journal of Mathematics 21 (2017), 599-630 (with L. Fresse).

67. On ideals in $U(sl(\infty))$, $U(o(\infty))$, $U(sp(\infty))$. In: Representation theory - current trends and perspectives, EMS Series of Congress Reports, European Mathematical Society (EMS), 2016, 565-602 (with A. Petukhov).

68. Ordered tensor categories and representations of the Mackey Lie algebra of infinite matrices, Algebras and Representation Theory 22 (2019), 249-279 (with A. Chirvasitu).

69. Real group orbits on flag ind-varieties of $\mathrm{SL}(\infty, \mathbb{C})$, In: Lie Theory and Its Applications in Physics, Springer Proceedings in Mathematics and Statistics, vol. 191 (2016), 111-135 (with M. V. Ignatyev and J. A. Wolf).

70. On the categories of admissible $\big(\mathfrak{g}, \mathrm{sl}(2)\big)$-modules,Transformation Groups 23(2) (2018), 463-489(with V. Serganova and G. Zuckerman).

71. Primitive ideals of $U(\mathfrak{sl}(\infty))$, Bulletin LMS 50 (2018), 443-448 (with A. Petukhov).

72. Ind-varieties of generalized flags: a survey of results, arXiv:1701.08478, preprint 2016 (with M. Ignatyev).

72'. Ind-varieties of generalized flags: a survey of results. In " Itogi nauki i techniki, series Modern mathematics and its applications", Russian Academy of Sciences 2018, Topic Surveys 147, 3–50, arXiv:1701.08478 (with M. Ignatyev), Russian version.

73. Decomposition of cohomology of vector bundles on homogeneous ind-spaces, C. R. Acad. Sci. Bulg. 70:7 (2017), 907-916 (with E. Hristova).

74. Orbit duality in ind-varieties of maximal generalized flags, Transactions of the Moscow Mathematical Society, 2017, 131-160 (with L. Fresse).

74'. Orbit duality in ind-varieties of maximal generalized flags, Transactions of the Moscow Mathematical Society 78 (2017), no 1(with L. Fresse), Russian version, arXiv:1704.03671.

75. Representation categories of Mackey Lie algebras as universal monoidal categories, Pure and Applied Mathematics Quarterly 13 (2017), 77-121 (with A. Chirvasitu).

76. Primitive ideals of $U(\mathfrak{sl}(\infty))$ and the Robinson-Schensted algorithm at infinity, arXiv:1801.06692, to appear in Representation of Lie Algebraic Systems and Nilpotent orbits, Progress in Mathematics, Birkhauser (with A. Petukhov).

77. Integrable $\mathfrak{sl}(\infty)$-modules and category $\mathcal{O}$ for $\mathfrak{gl}(m|n)$, Journal LMS, first publishe online 14 September, 2018, doi: 10.1112/jlms.12176, arXiv:1712.00664 (with C. Hoyt and V. Serganova).

78. On an infinite limit of BGG categories $\mathcal{O}$, Moscow Mathematical Journal, to appear; arXiv:1802.06343 (with K. Coulembier).

79. On existence of infinite-dimensional generalized Harish-Chandra modules, Sao-Paulo Jour. Math. 12(2018), 290-294 (with G. Zuckerman).

80. Simple bounded weight modules of $\mathfrak{sl}(\infty)$, $\mathfrak{o}(\infty)$, $\mathfrak{sp}(\infty)$, arXiv:1807.01899, preprint 2018 (with D. Grantcharov).

81. Large annihilator category $\mathcal{O}$ for $\mathfrak{sl}(\infty)$, $\mathfrak{o}(\infty)$, $\mathfrak{sp}(\infty)$, Journal of Algebra 532 (2019), 249-279 (with V. Serganova).

82. Examples of automorhism groups of ind-varieties of generalized flags, Journal of Geometry and Symmetry in Physics 50(2018),71-77 (doi:10.7546/jgsp-50-2018-71-77).

Books edited

1. V. Chari, I. B. Penkov - editors. Modular Interfaces: Modular Lie Algebras, Quantum Groups, and Lie Superalgebras. "Studies in Advanced Mathematics", vol. 4, 1997, AMS.

2. A. Joseph, A. Melnikov, I. Penkov - editors. Highlights in Lie Algebraic Methods. "Progress in Mathematics", vol. 295, 2011, Birkhauser.

3. A. Huckleberry, I. Penkov, G. Zuckerman - editors. Lie Groups: Structure, Actions, and Representations. "Progress in Mathematics", vol. 306, 2013, Birkhauser.